

Math 411 Summer 2025

Midterm 2

August 22, 2025

8:30 - 9:30 AM

Name: _____

Student ID: _____

Instructions:

1. You have 60 minutes to finish the exam.
2. You may only work on your own.
3. Please try to write as legible as you can.
4. Do not write on the back of any page.
5. Please provide as many detail as you can.
6. Write down and sign your honor statement.

Question:	1	2	Total
Points:	10	15	25
Score:			

Please note the final exam will have 3 problems and 30 points in total.

1. Let K be a field and $K[x]$ be the polynomial ring with coefficients in K . Determine if the following is true or not and give an explanation/counterexample.
 - (a) (2 points) If a polynomial in $K[x]$ has a root in K , then it is reducible.
 - (b) (2 points) If $f(x) \in \mathbb{Z}[x]$ is irreducible in $\mathbb{Q}[x]$, then $f(x)$ is irreducible in $\mathbb{Z}[x]$.
 - (c) (2 points) Every Unique factorization domain is a Principle ideal domain.
 - (d) (2 points) Every maximal ideal in $K[x]$ is generated by some $f(x) \in K[x]$ of degree 1.
 - (e) (2 points) In K , irreducible elements are prime.

Proof. 1. No. Because the polynomial may be of degree 1. For instance, $x - 1$ has a root in $\mathbb{Q}$ and it is irreducible.

2. No. It must also be primitive. For instance, $2x - 2$ is of degree 1 in $\mathbb{Q}[x]$ and hence is irreducible in $\mathbb{Q}[x]$. But $2x - 2 = 2(x - 1)$ where $x - 1$ and 2 are irreducible in $\mathbb{Z}[x]$.

3. No. Consider $\mathbb{Z}[x]$ where $(2, x)$ is not principle.

4. No. It can be generated by any irreducible polynomials which can be of higher degree. For instance, $x^4 - 2$ which is irreducible by the Eisenstein criterion.

5. This is true because there are no irreducible elements not prime elements in K . The statement is vacuously true. $\square$

2. Let K be a field. Let $f = f(x) \in K[x]$. Define $V(f) = \{p \in K : f(p) = 0\} \subseteq K$. Let $\gamma \in K$. Define $I(\gamma) = \{f \in K[x] : f(\gamma) = 0\} \subseteq K[x]$.

For the rest of the question, assume $K = \mathbb{R}$. Show the following.

- (a) (1 point) Let $g(x) = x^4 - \frac{5}{2}x^3 - x^2 - 2x + \frac{3}{2}$. Is $g(x)$ irreducible in $\mathbb{R}[x]$?
- (b) (2 points) Let $g(x) = x^4 - \frac{5}{2}x^3 - x^2 - 2x + \frac{3}{2}$. What is $V(g) \subseteq \mathbb{R}$?
- (c) (2 points) Give two examples of $f = f(x) \in K[x]$ with different degrees such that $V(f) = \emptyset$.
- (d) (2 points) Let $\gamma \in K$. Show $I(\gamma)$ is an ideal in $K[x]$.
- (e) (2 points) Let $f = f(x) \in K[x]$ be of positive degree. Show that if $V(f) \neq \emptyset$, then (f) is contained in some maximal ideal in $K[x]$.
- (f) (2 points) Let $f = f(x) \in K[x]$ be of positive degree. Show that if $V(f) = \emptyset$, then (f) is contained in some maximal ideal in $K[x]$.
- (g) (1 point) Show every proper ideal in $K[x]$ is contained in some maximal ideal.
- (h) (1 point) Is the maximal ideal in the previous question unique?
- (i) (2 points) Is it true that every proper ideal in $\mathbb{Z}[x]$ is contained in some maximal ideal?

Proof. 1. First we write

$$g(x) = \frac{1}{2}(2x^4 - 5x^3 - 2x^2 - 4x + 3)$$

By the rational root test, the rational roots are $\pm 1, \pm 3, \pm 1/2, \pm 3/2$. We may check that $1/2$ is a root. Then $2x - 1 \mid g(x)$, which implies $g(x)$ is reducible in $\mathbb{R}[x]$.

2. By part 1, we know $1/2 \in V(g)$. Use long division of polynomial, we get

$$g(x) = \frac{1}{2}(2x - 1)(x^3 - 2x^2 - 2x - 3).$$

Any rational roots must be of the form $\pm 1, \pm 3$. We may compute that 3 is a rational root. Then we may further factor

$$g(x) = \frac{1}{2}(2x - 1)(x - 3)(x^2 + x + 1).$$

Now the last factor is irreducible in $\mathbb{Z}_2[2]$ which implies it is irreducible in $\mathbb{Q}[x]$. It is also irreducible in $\mathbb{R}[x]$ as the roots will involve cube root of unity. So we conclude that $V(g) = \{\frac{1}{2}, 3\}$.

3. Consider $x^2 + 1$ and 2.

4. First, the constant zero polynomial lies in $I(\gamma)$. Second, if $f(\gamma) = g(\gamma) = 0$, then $(f + g)(\gamma) = f(\gamma) + g(\gamma) = 0$. Finally, if $f(\gamma) = 0$ and $h(x) \in K[x]$, then $h(\gamma)f(\gamma) = h(\gamma) \cdot 0 = 0$.

Or you can say the evaluation map at γ is a ring homomorphism, and the kernel is $I(\gamma)$ which must be an ideal.

5. Suppose $V(f) \neq \emptyset$. This implies $f(x)$ admits a root $p \in K$. Therefore $(x-p)h(x) = f(x)$ for some $h(x) \in K[x]$. It follows that $f(x) \in (x-p)$. The latter is a maximal ideal because K is a field and $K[x]$ is a PID. So any irreducible polynomial generates a maximal ideal. Here $x-p$ is of degree 1 and hence irreducible.
6. Suppose $V(f) = \emptyset$. So $f(x)$ has no roots in K . But since $K[x]$ is a UFD, $f(x) = p_1(x) \cdots p_n(x)$ for $p_i(x)$ irreducible. In particular, it follows $f(x) \in (p_i(x))$ which is a maximal ideal since $p_i(x)$ is irreducible.
7. Recall $K[x]$ is a PID. So any proper ideal has the form $(f(x))$ for some $f(x)$ of positive degree. By Part e and f, regardless $V(f)$ is empty or not, $f(x)$ is contained in some maximal ideal $\mathfrak{m}$. This follows that $(f(x)) \subseteq \mathfrak{m}$.
8. No. It is not unique. For instance, consider $x(x-1)$ which lies in maximal ideals (x) and $(x-1)$.
9. Yes. The difficulty is that $\mathbb{Z}[x]$ is no longer a PID. But luckily Noetherianness comes to rescue. Take any ideal $I \subset \mathbb{Z}[x]$. If I is maximal, then we are done. Suppose I is not maximal. Then there exists an element $x_1 \notin I$ such that $I = I_0 \subseteq (I, x_1) = I_1 \neq \mathbb{Z}[x]$. If I_1 is maximal, then we are done. If not, we repeat the process and find an element $x_2 \notin I_1$ such that $I = I_0 \subseteq (I, x_1) = I_1 \subset (I, x_1, x_2) = I_2 \neq \mathbb{Z}[x]$. This process induces an ascending chain of ideals in $\mathbb{Z}[x]$ which is Noetherian. So the chain must stabilize. Then the last distinct term in the chain is a maximal ideal.

This result is true for any commutative ring and is generally proved by what is called the Zorn's lemma. But for PID and Noetherian rings, one can avoid the Zorn's lemma. $\square$